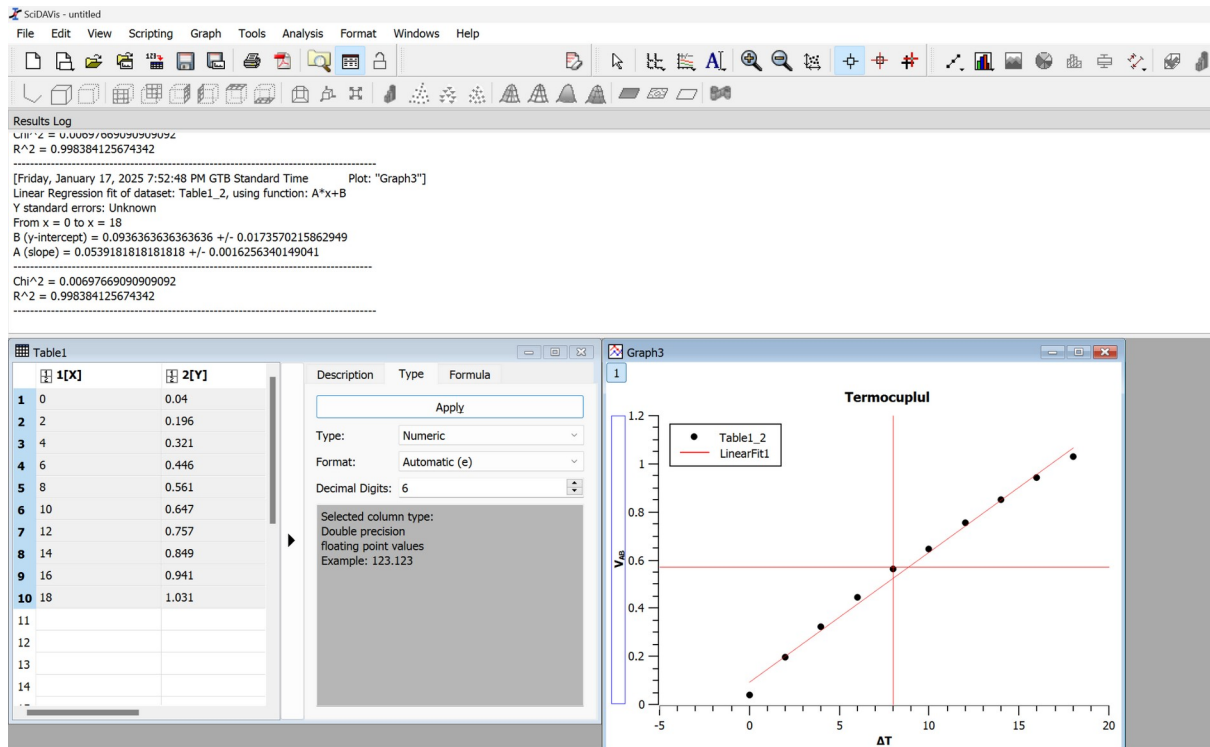


Referate laborator MT

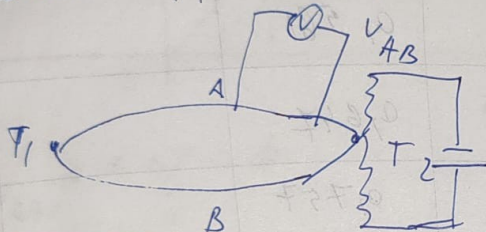
Lab1 - termocuplul



Termocuplul

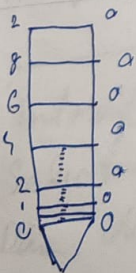
- senzor de temperatură
- se bazează în fct. reef. Seebeck

Dacă $T_1 \neq T_2$, atunci (7) V_{AB}



$$V_{AB} = \alpha \cdot \Delta T$$

→ forța termoelectrică
at. de material?



$$(0^\circ) + \Delta = \alpha(T) \Delta$$

N_4	$t_1(^{\circ}\text{C})$	$t_2(^{\circ}\text{C})$	$\Delta T(\text{K})$	$V_{AB}(\text{mV})$	$S(\text{mV}/\text{K})$
1	24	25	0	0,09	
2		26	2	0,136	
3		28	4	0,321	
4		30	6	0,446	
5		32	8	0,561	
6		34	10	0,647	
7		36	12	0,757	
8		38	14	0,843	
9		40	16	0,941	
10		42	18	1,031	0,059

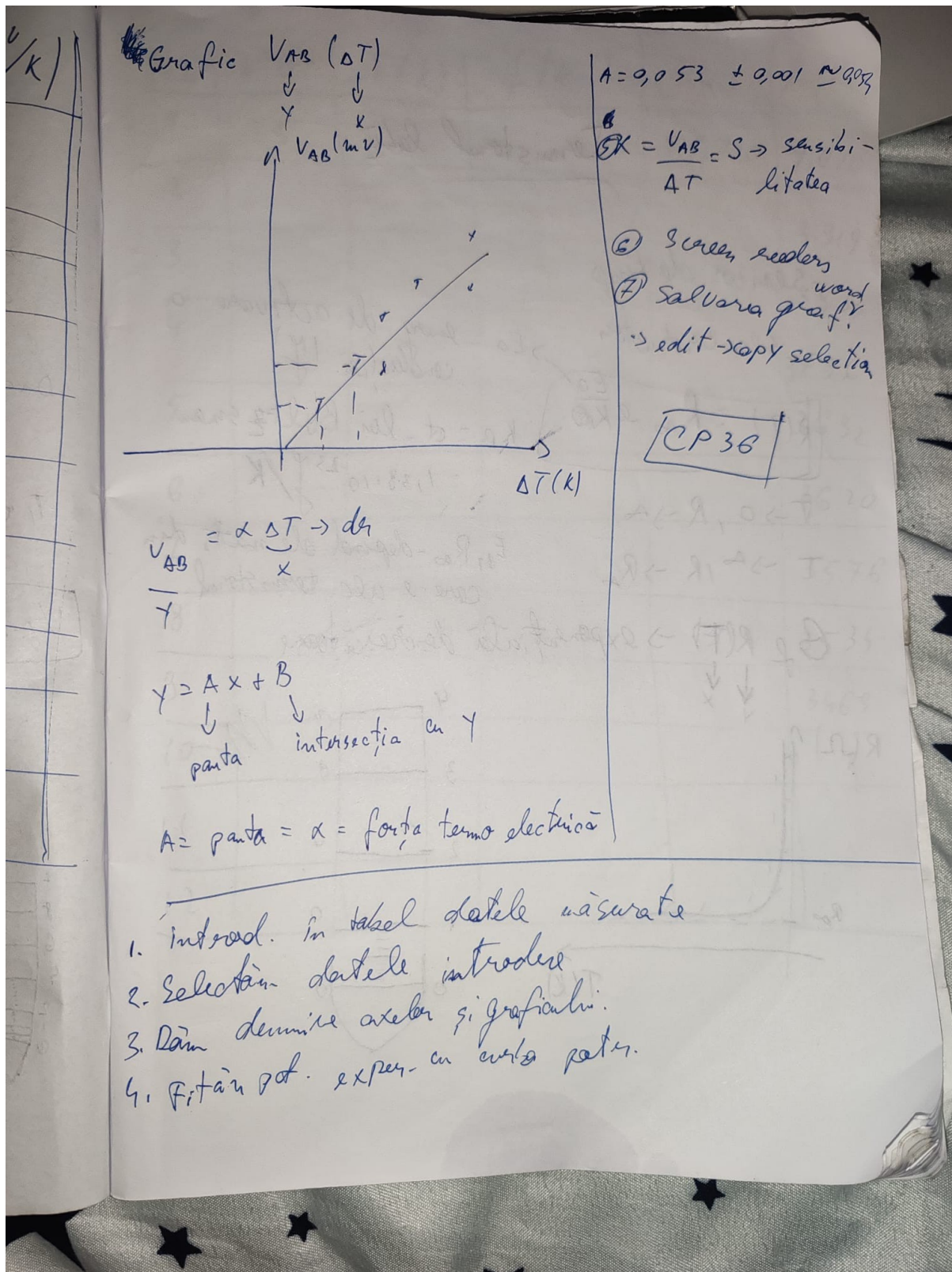
$$T(\text{K}) = t(^{\circ}\text{C}) + 273,15$$

$$T_2(\text{K}) = t_2(^{\circ}\text{C}) + 273,15$$

$$T_1(\text{K}) = t_1(^{\circ}\text{C}) + 273,15 \quad (-)$$

$$\Delta T(\text{K}) = \Delta t(^{\circ}\text{C})$$

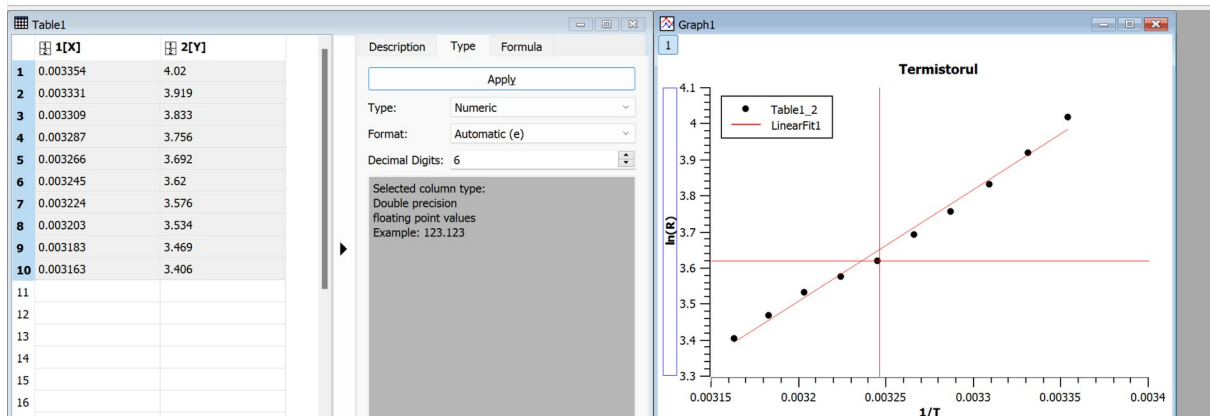
- 1.
- 2.
- 3.
- 4.



Results Log

[Friday, January 17, 2025 8:08:27 PM GTB Standard Time Plot: "Graph1"]
Linear Regression fit of dataset: Table1_2, using function: $A \cdot x + B$
Y standard errors: Unknown
From $x = 0.003163$ to $x = 0.003354$
 B (y-intercept) = $-6.3810027141256 \pm 0.364020403489069$
 A (slope) = $3,090.28180995719 \pm 111.763202229149$

 $\chi^2 = 0.00370018790931295$
 $R^2 = 0.999972785808148$



Termistorul labu

- senzor de timp

- semiconductor

E_a = energ. de activare a conductiei (J)

$$R(T) = R_{\infty} e^{\frac{E_a}{k_B T}}$$

k_B = ct. lui Boltzmann

$$= 1,38 \cdot 10^{-23} \text{ J/K}$$

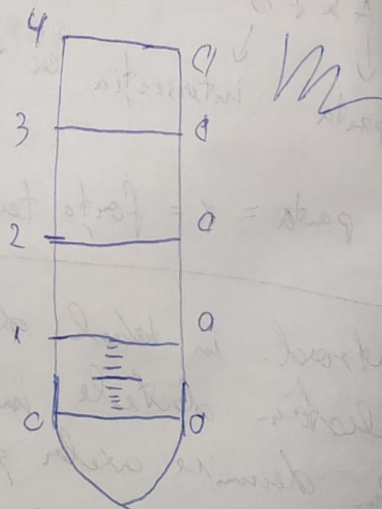
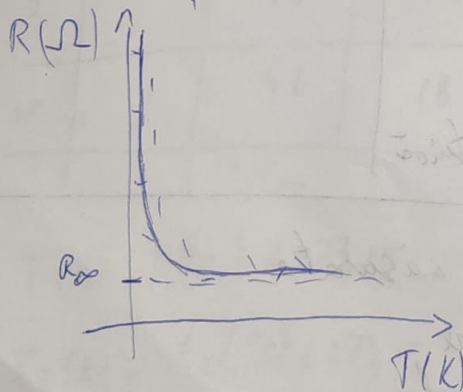
$T \rightarrow 0, R \rightarrow \infty$

$T \rightarrow \infty, R \rightarrow R_{\infty}$

E_a, R_{∞} - depind de mat. din care e alc. termistorul

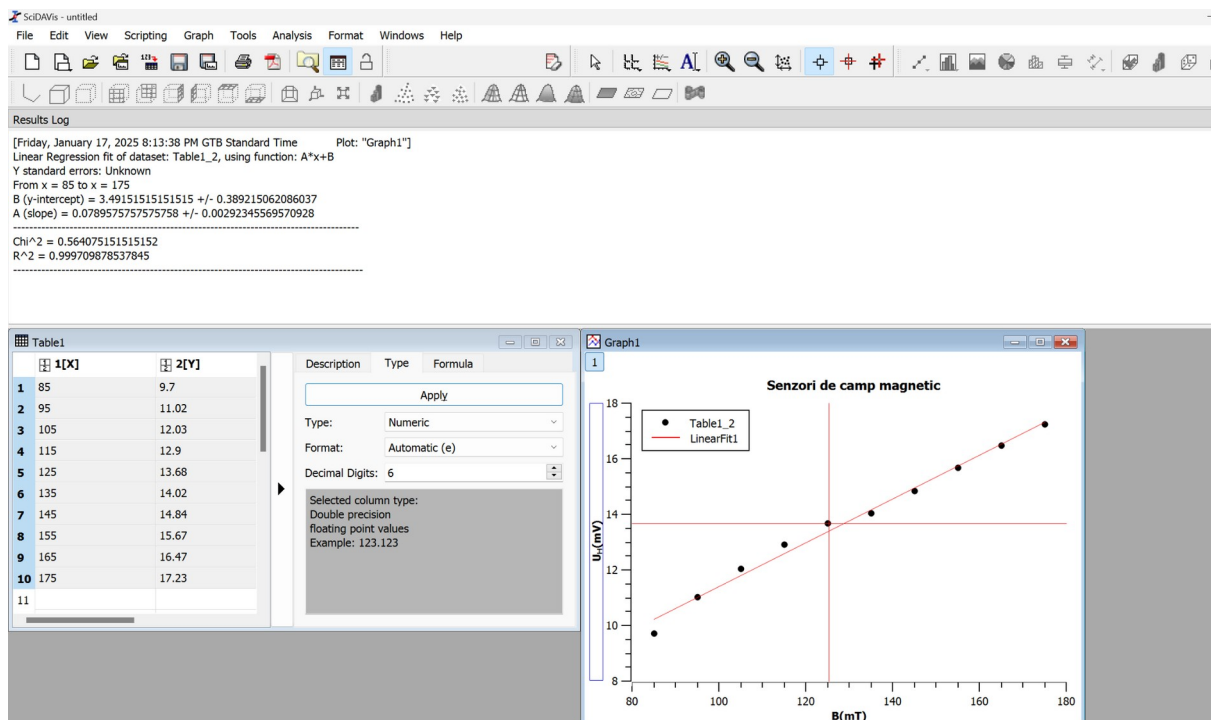
Gf $R(T) \rightarrow$ exponentială descrescătoare

↓
↓
x



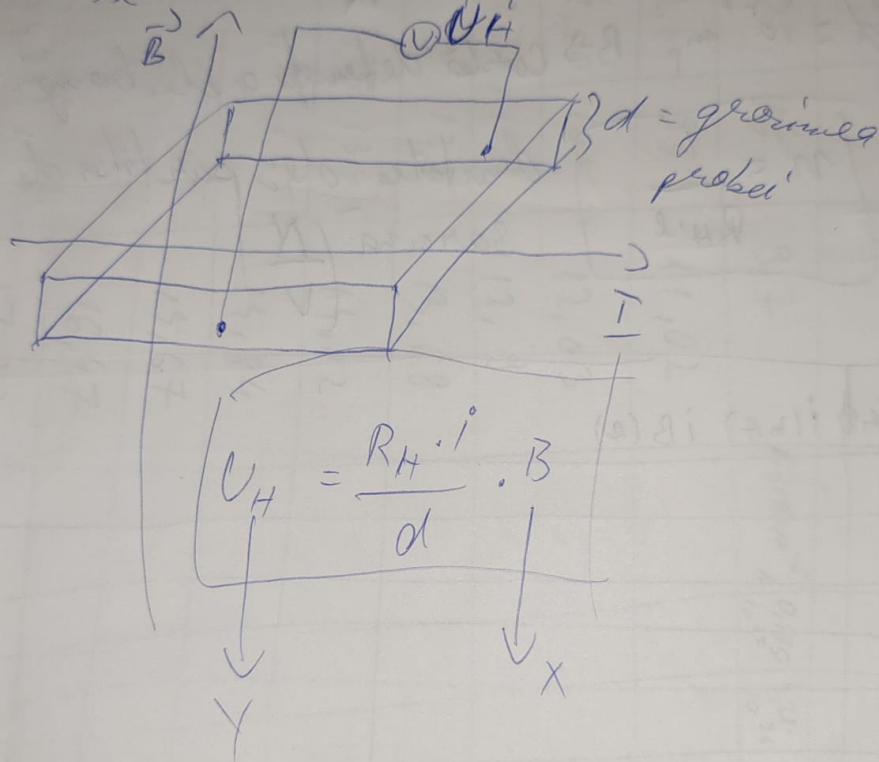
N ₆	t ^{°C}	T(K)	$\frac{1}{T} (K^{-1})$	R ₅₀	ln(R)
1	25	298.15	0,003354	55,710	4,020
2	27	300.15	0,003331	50,370	3,919
3	29	302.15	0,003309	46,220	3,833
4	31	304.15	0,003287	42,812	3,756
5	33	306.15	0,003266	40,130	3,692
6	35	308.15	0,003245	37,550	3,620
7	37	310.15	0,003224	35,753	3,576
8	39	312.15	0,003203	34,270	3,535
9	41	314.15	0,003183	32,130	3,468
10	43	316.15	0,003163	30,153	3,406
11					
12					

Lab 3 - senzor de camp magnetic



Senzor de câmp magnetic

- se bazează în func. pe ef Hall.



$$U_H(B) = \text{func. de } g \cdot I, \text{ Gf decapata}$$
$$\text{panta} = \frac{R_H \cdot I}{d} \quad (\text{panta } \downarrow A)$$

$$R_H = \frac{d \cdot A}{i} \quad \text{et Hall}$$

$$I = \frac{U}{R}$$

$d = 10^{-7} \text{ m}$ $B \rightarrow$ curba de func. a electroni

$$n = \frac{1}{R_H \cdot d} \quad \text{densitate de purtători de sarcină} \left(\frac{N}{V} \right)$$

N_A $i (mA)$ $i_B (A)$

1

2

3

4

5

N_A	$i(mA)$	$ B(A) $	$B(mT)$	$V_H(mV)$	R_H	n_1	n_2
1	6	2	85	9.7	$1.315 \cdot 10^{-6} A$	$0.975 \cdot 10^{25}$	$4.21 \cdot 10^{24}$
2		2.25	95	11.02			
3		2.50	105	12.03			
4		2.75	115	12.9			
5		3	125	13.68			
6		3.25	135	14.02			
7		3.50	145	14.84			
8		3.75	155	15.67			
9		4	165	16.47			
10		4.25	175	17.23			

$t = 0.08805$

Calc. er.

1) abs.

$$\Delta n = n - \bar{n}$$

$$\Delta n_3 = n_3 - \bar{n} = 0,54 \cdot 10^{24}$$

2) er. relativă

$$\Sigma \% = \frac{\Delta n}{n}$$

$$\Sigma_1 = \frac{\Delta n_3}{n_3} = 0,11368 \approx 11\%$$

3) Er. standard a mediei aritmet.

$$\sigma_n = \frac{\sum_{i=1}^3 (\Delta n_i^2)}{N(N-1)}$$

$$\underbrace{N(N-1)}_6$$

$$\frac{\sqrt{\Delta n_1^2 + \Delta n_2^2 + \Delta n_3^2}}{6}$$

$$N=3$$

$$\frac{10^{24}}{6}$$

$$\cdot \sqrt{0,18^2 + 0,35^2 + 0,54^2}$$

$$\boxed{0,11 \cdot 10^{24}}$$

Lab4 - Dete. TEM a unei surse prin metoda compensarii

Det. TEM a unei surse
prin met. compensații

Nr	I_1			R_1	E_x	$\langle E_x \rangle$
	div	$K(A/div)$	A	R	V	V
1	75	10^{-5}	0,00075	4000	3	2,98
2	71	10^{-5}	0,00071	4200	2,98	
3	68	10^{-5}	0,00068	4400	2,98	
4	65	10^{-5}	0,00065	4600	2,99	
5	62	10^{-5}	0,00062	4800	2,976	
6	60	10^{-5}	0,00060	5000	3	

$E, e, E = t_{em}$

$E_0 = t_{em} \text{ etalon}$

$E_x = t_{em} \text{ necunoscută}$

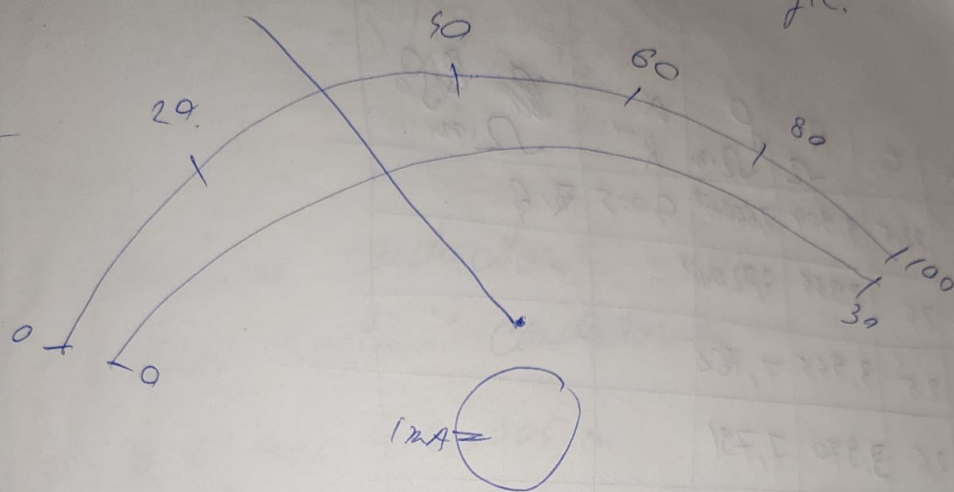
$$E_x = E_0 \cdot \frac{R_x}{R}$$

$$E_x = i_x R_x + i_1 R_1$$

Dacă $i_x = 0 \Rightarrow E_x$

9,01; - 0,00

Măsurătorii cu multimetru analogic.



$$1 \text{ mA} \text{ --- } 100 \text{ div}$$

$$x \text{ --- } 10 \text{ div}$$

$$x = \frac{1 \text{ mA}}{100} = 10^{-5} \text{ A}$$

$k = \text{ct. aparatului}$

$$k = 10^{-5}$$

$$\Rightarrow \text{m de } \frac{\text{A}}{\text{div}} = \text{ca } 10^{-5} \text{ A.}$$

$$1 \text{ mA} \text{ --- } 30 \text{ div}$$

$$x_2 \text{ --- } 1 \text{ div}$$

$$x_2 = \frac{1 \text{ mA}}{30} = \frac{1}{3} \cdot 10^{-3} \text{ A} = 0,33 \cdot 10^{-3} \text{ A / K}$$

Termorezistorul

Nr	t_{oc}	R Ω	ρ $\Omega \cdot m$	α K^{-1}	R_0 R_0 Ω
1	22,5	3,304	$7,656 \cdot 10^{-8}$	0,005	6 6
2	24	3,339	$7,720 \cdot 10^{-8}$		
3	25	3,355	7,762		
4	26	3,370	7,791		
5	27	3,395	7,840		
6	28	4,003	7,868		
7	29	4,026	7,90		
8	30	4,043	7,934		
9	31	4,065	7,978		
10	32	4,079	8,005		

- mat. conduct. : metal.

- Senzor de temp. :

$$R = R_0(1 + \alpha t)$$

t_{temp} - fct. q $\bar{t}$ $\Rightarrow R$ creste lin

$$\rho = \rho_0 (1 + \alpha t) - \text{rezistivitatea el } (\Omega \cdot m)$$

$$\boxed{R = \frac{\rho l}{S}} \Rightarrow \boxed{\rho = \frac{R \cdot S}{l}} \quad 7,656 \cdot 10^{-8}$$

l - lungimea conductorului = 10 m

S - aria secțiunii conductorului

$$\phi = 0,5 \text{ mm} = 0,0005 \text{ m}$$

$$S = \pi r^2 = \pi \left(\frac{\phi}{2} \right)^2 = \frac{\pi \phi^2}{4} = \frac{\pi (0,5)^2}{4}$$

$$= \frac{0,25 \pi}{4} = \frac{\pi \cdot 0,0005^2}{4}$$

$$= 1,9625 \cdot 10^{-7} \text{ m}^2$$

Grafic $R(t), \rho(t)$

$\downarrow \quad \downarrow$
Y X

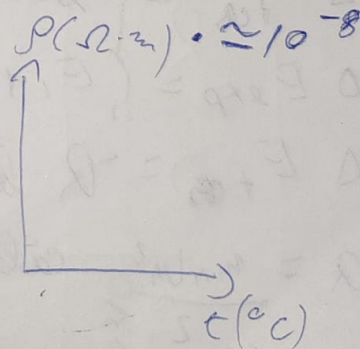
$$R = R_0 + R_0 \alpha t$$

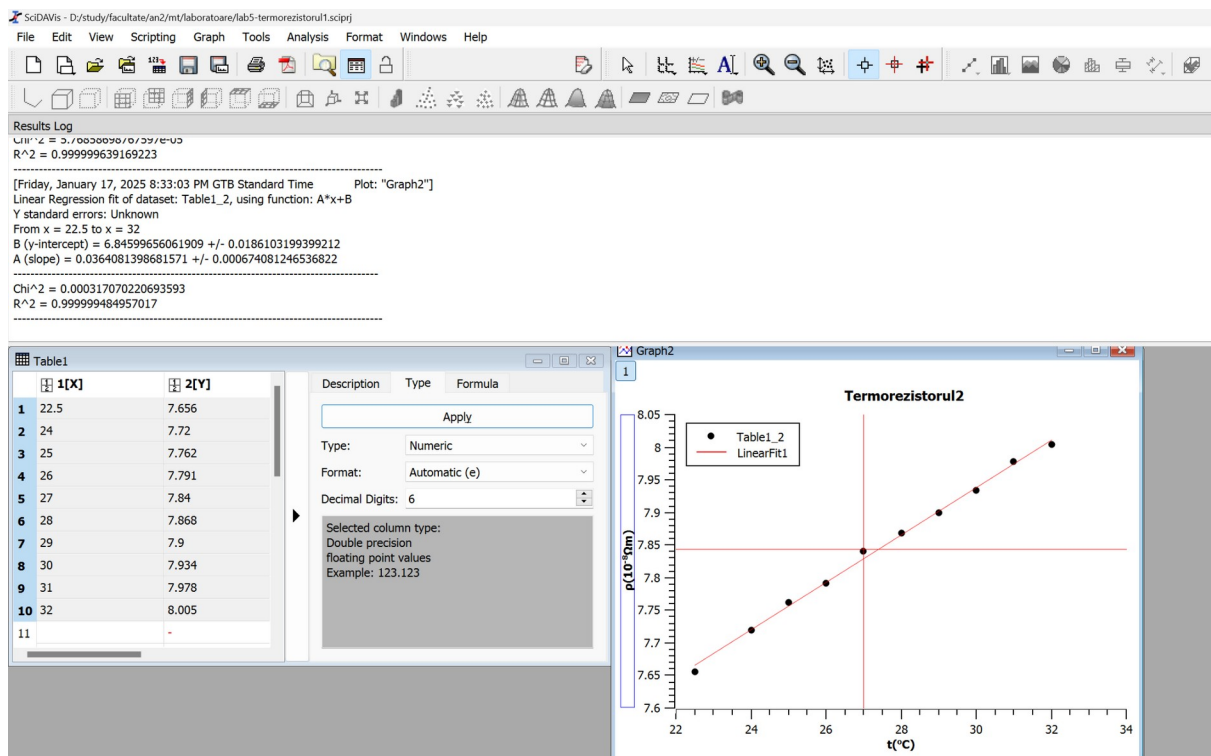
$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
Y B A X
panta

$B = R_0$ - rezist. el la 0°C

$$A = R_0 \alpha \text{ (panta)} \Rightarrow \alpha = \frac{A}{R_0} = \frac{A}{B}$$

$\hookrightarrow$ coef. termic al rezistivității





Lab6 ciocniri

Sistem. de achiziție
de date pt. studiu ciocnirilor

I $m_1 = m_2$

II $m_1 \gg m_2$

III $m_1 \ll m_2$

$v_1 =$ înainte de c.

$v_1' =$ după c.

$p = mv$

$p_1 = m_1 v_1$ - înainte de c.

$p_1' = m_1 v_1'$ - după c.

$$p_1 + p_2 = p_1' + p_2'$$

$$E = \frac{mv^2}{2}$$

$$E_1 + E_2 = E_1' + E_2' \quad \text{ciocn. pe el.}$$

$$\Delta E_{\text{tes}} = 0 - \text{la ciocnire perfect el.}$$

$$\Delta E_{\text{exp}} = (E_1' + E_2') - (E_1 + E_2)$$

$$\Delta E_{\text{tes}} = -Q \text{ la ciocn. plastică.}$$

$$Q = \frac{m_A v_A^2}{2} \quad \text{căldură}$$

$$m_A = \frac{m_1 m_2}{m_1 + m_2} \quad \text{- masa redusă a sist.}$$

$$v_A = v_{\text{CM}} \text{ relativă a cap.}$$

$$v_A = |v_1 - v_2|$$

Nr crt.	Compu	m	V	V'	P	P'	E	E'	P ₁ P ₂	P ₁ P ₀	E ₁ E ₂	E ₁ E ₂	ΔE _{exp}	ΔE _{teor}
1		kg	m/s	m/s	Ns	Ns	f	f	Ns	Ns	f	f	f	f
1		0,213	0,739	0,789	0,170	0	0,06	0	0,15975	0,0382	0,06	0,05	0,02	0
2	elastica	0,213	0,739	0,789	0	0,170	0	0,06	0,15975	0,0382	0,06	0,05	0,02	0
1		0,213	0,703	0,375	0,15975	0,0787	0,06	0,01	0,15975	0,1915	0,06	0,05	0,02	0
2	plastică	0,213	0	0,297	0	0,0686	0	0,003	0,15975	0,1915	0,06	0,05	0,02	0

$$E = \frac{mV^2}{2} \Rightarrow E = \frac{0,213 \cdot 0,750^2}{2} = 0,0050105$$

$$-Q = 0,00310$$

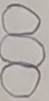
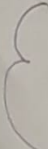
Lab7 - masurari cu osciloscopul

Senzorii tactili

Def. func. unei tens. alternative
cu ajutorul osciloscopului

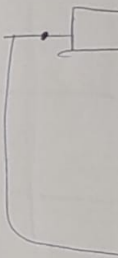
$$u = U_{max} \cos(\omega t + \phi_0)$$

$$\omega = 2\pi f$$

Nr. Fig. Lissajous	D_x / Hz	n_x	n_y	n	D_y / Hz	$\bar{D}_y / Hz$
1 		2	2	1	45,6	
2 		3	4	$\frac{3}{4}$	45,525	
3 		2	3	$\frac{2}{3}$	45,078	45,1
4 		1	2	$\frac{1}{2}$	45,6	
5 		2	6	$\frac{1}{3}$	45,21	
6 		1	4	$\frac{1}{4}$	45,65	

$n_x, n_y =$
Lissaj
 $n = \frac{m}{n}$

$D_y =$



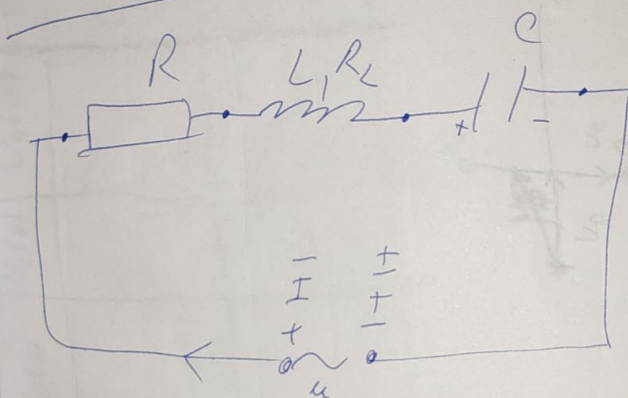
$u =$
 $u =$
 U_{max}
 $U =$
 U_{max}
 $D = \frac{1}{T}$

$n_x, n_y = \text{nr. de intersecții dintre o fig. liessajor și axele } x, y.$

$$n = \frac{n_x}{n_y} =$$

$$\partial_y = n \partial_x$$

$\gamma(H_z)$



95,44

$$u = U_{\max} \sin(\omega t + \varphi)$$

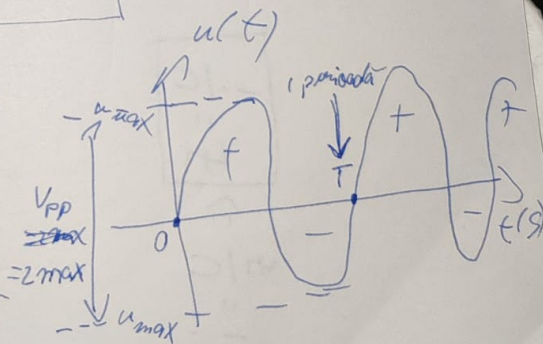
$u = \text{tens. instantanee}$

$U_{\max} = \text{val. max. a tens}$

$U = U_{ef} = \text{tens efectivă}$

$$U_{\max} = \sqrt{2} U$$

$$\omega = \frac{1}{T} [H_z]$$



$$f_{\text{prize Ro}} = 50 \text{ Hz (complete oscillation/seconds)}$$

$$1 \text{ KHz} = 10^3 \text{ Hz}$$

$$1 \text{ MHz} = 10^6 \text{ Hz (1 million)}$$

V_{PP} = rms peak to peak

$$V_{PP} = 2 V_{\text{max}}$$

$$V_{\text{RMS}} = V_{\text{ef}}$$

Results Log

K² = 0.903 / 7441001000

[Friday, January 17, 2025 8:46:21 PM GTB Standard Time Plot: "Graph1"]

Polynomial fit of dataset: Table1_2, using function: $a_0 + a_1x + a_2x^2$

Y standard errors: Unknown

From x = 0 to x = 2

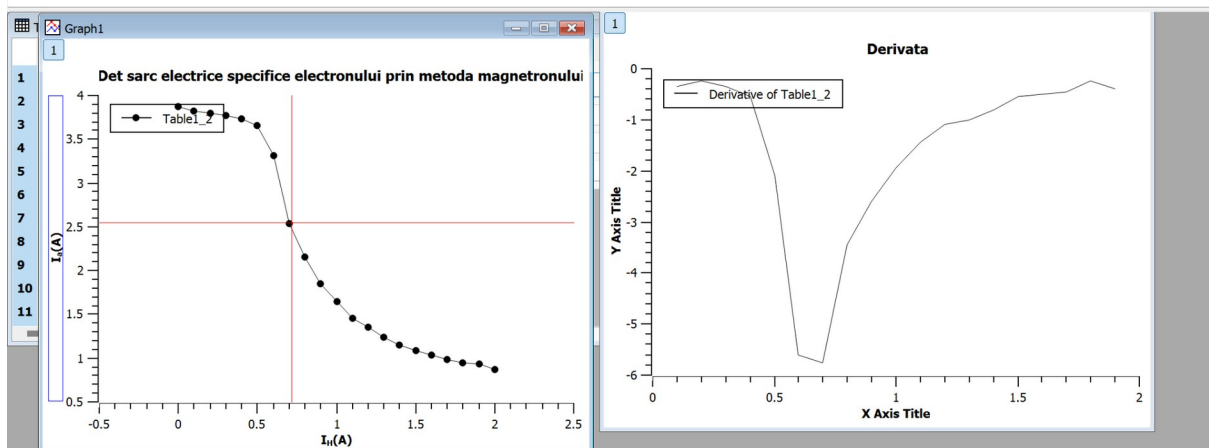
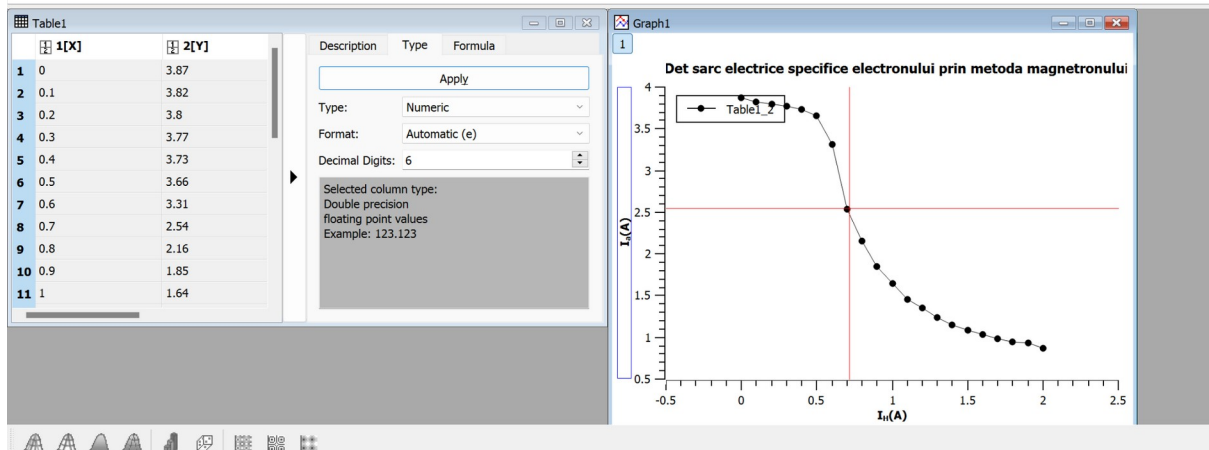
$a_0 = 4.4226990400903 \pm 0.188558088945289$

$a_1 = -3.1905379060299 \pm 0.43691347826416$

$a_2 = 0.673645576391571 \pm 0.210911042018627$

Chi^2 = 1.79618904335939

R^2 = 0.985774224884888



Lab10

SciDAVis - untitled

File Edit View Scripting Graph Tools Analysis Format Windows Help

Results Log

[Friday, January 17, 2025 8:53:06 PM GTB Standard Time Plot: "Graph1"]
 Linear Regression fit of dataset: Table1_2, using function: $A \cdot x + B$
 Y standard errors: Unknown
 From $x = 1$ to $x = 2.8$
 B (y-intercept) = $0.3187878787859 \pm 0.10667054056326$
 A (slope) = $45.28484848485 \pm 0.0537398419726293$

 $\chi^2 = 0.07624242424197$
 $R^2 = 0.999999062761463$

